

AN ELEMENTARY INTRODUCTION TO MICROECONOMICS



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The business meaning matters as much as the arithmetic.

This section builds the idea of owners, managers and agency theory before moving to application.

Agency cost. A cost the principal bears to reduce the agency problem or from the loss it still causes..

The term agency cost gives the chapter a precise name for this part of the problem.

Incentive. A reward, such as a bonus, that encourages a person to act in a chosen way..

The term incentive gives the chapter a precise name for this part of the problem.

The central question in owners, managers and agency theory is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For wealth maximisation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

3.3 OWNERS, MANAGERS AND AGENCY THEORY

The purpose here is to make owners, managers and agency theory concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Agency cost** means A cost the principal bears to reduce the agency problem or from the loss it still causes. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Incentive** means A reward, such as a bonus, that encourages a person to act in a chosen way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For owners, managers and agency theory, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the goal of the firm and agency theory. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the

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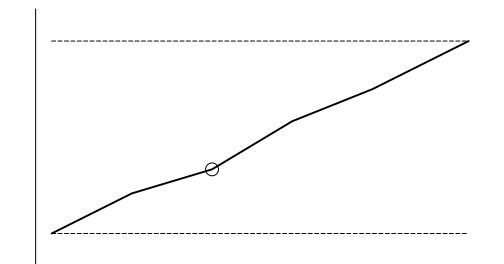


FIG. 3.1. The Goal of the Firm and Agency Theory figure 1

3.1 PROFIT MAXIMISATION AND ITS LIMITS

This section builds the idea of profit maximisation and its limits before moving to application.

Profit maximisation. Aiming to make the largest profit possible, usually within one year..

The term profit maximisation gives the chapter a precise name for this part of the problem.

Wealth maximisation. Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is..

The term wealth maximisation gives the chapter a precise name for this part of the problem.

Shareholder. A person who owns part of a company by holding its shares..

The term shareholder gives the chapter a precise name for this part of the problem.

Share price. The price at which one share of a company can be bought or sold today..

The term share price gives the chapter a precise name for this part of the problem.

The central question in profit maximisation and its limits is what changes when the underlying decision changes.

The purpose here is to make profit maximisation and its limits concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profit maximisation** means Aiming to make the largest profit possible, usually within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Wealth maximisation** means Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Shareholder** means A person who owns part of a company by holding its shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Share price** means The price at which one share of a company can be bought or sold today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches An Elementary Introduction to Microeconomics one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

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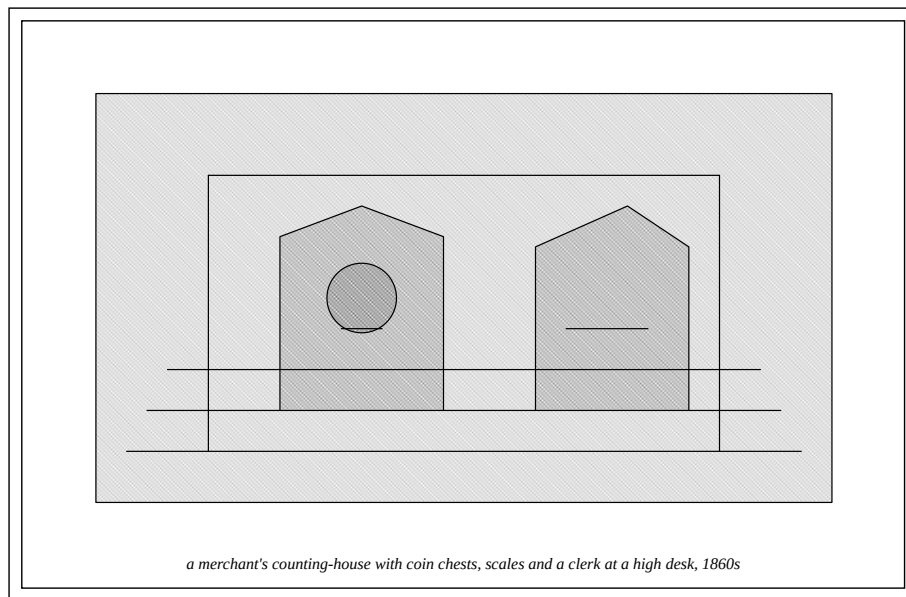
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CHAPTER I

What Finance Is

A practical question about what finance is



This chapter introduces what finance is for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A calculation can be correct and still answer the wrong question. For finance in plain words, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

1.2 FINANCE BESIDE ITS NEIGHBOURING SUBJECTS

The purpose here is to make finance beside its neighbouring subjects concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating efficiency** means How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sole proprietorship** means A business owned and run by one person, who keeps all profit and answers for all debts. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Partnership** means A business owned by two or more people who agree to run it together and share its profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For finance beside its neighbouring subjects, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

The business meaning matters as much as the arithmetic.

This section builds the idea of finance beside its neighbouring subjects before moving to application.

Operating efficiency. How much useful output a business gets from the money and effort it uses.

The term operating efficiency gives the chapter a precise name for this part of the problem.

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts..

The term sole proprietorship gives the chapter a precise name for this part of the problem.

Partnership. A business owned by two or more people who agree to run it together and share its profit..

The term partnership gives the chapter a precise name for this part of the problem.

The central question in finance beside its neighbouring subjects is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

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This section builds the idea of decisions on the balance sheet before moving to application.

Balance sheet. A statement listing what a business owns and what it owes on one date..

The term balance sheet gives the chapter a precise name for this part of the problem.

Asset. Something a business owns that has value, such as cash, stock or a van..

The term asset gives the chapter a precise name for this part of the problem.

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill..

The term liability gives the chapter a precise name for this part of the problem.

Equity. What is left for the owners when liabilities are subtracted from assets..

The term equity gives the chapter a precise name for this part of the problem.

The central question in decisions on the balance sheet is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

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A calculation can be correct and still answer the wrong question. For what a financial manager does, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

2.3 DECISIONS ON THE BALANCE SHEET

The purpose here is to make decisions on the balance sheet concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Balance sheet** means A statement listing what a business owns and what it owes on one date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Asset** means Something a business owns that has value, such as cash, stock or a van. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Liability** means Money a business owes to others, such as a bank loan or an unpaid supplier bill. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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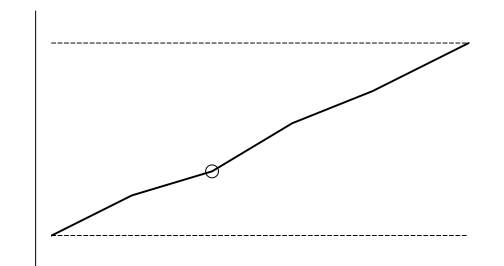


FIG. 2.1. *The Financial Manager's Decisions figure 1*

2.1 THE THREE BIG DECISIONS

The purpose here is to make the three big decisions concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial manager** means The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Investment decision** means A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financing decision** means A choice about where the money comes from: owners' own funds, a loan, or new owners. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Dividend** means A share of a company's profit paid out to its owners instead of being kept in the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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This section builds the idea of the three big decisions before moving to application.

Financial manager. The person who decides how a business finds, uses and protects its money.

The term financial manager gives the chapter a precise name for this part of the problem.

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money..

The term investment decision gives the chapter a precise name for this part of the problem.

Financing decision. A choice about where the money comes from: owners' own funds, a loan, or new owners..

The term financing decision gives the chapter a precise name for this part of the problem.

Dividend. A share of a company's profit paid out to its owners instead of being kept in the business..

The term dividend gives the chapter a precise name for this part of the problem.

The central question in the three big decisions is what changes when the underlying decision changes.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of What Finance Is is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

The Dutch East India Company (VOC), chartered 1602, sold shares to the public that traded in Amsterdam, an early company separating owners from managers. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

REVIEW QUESTIONS

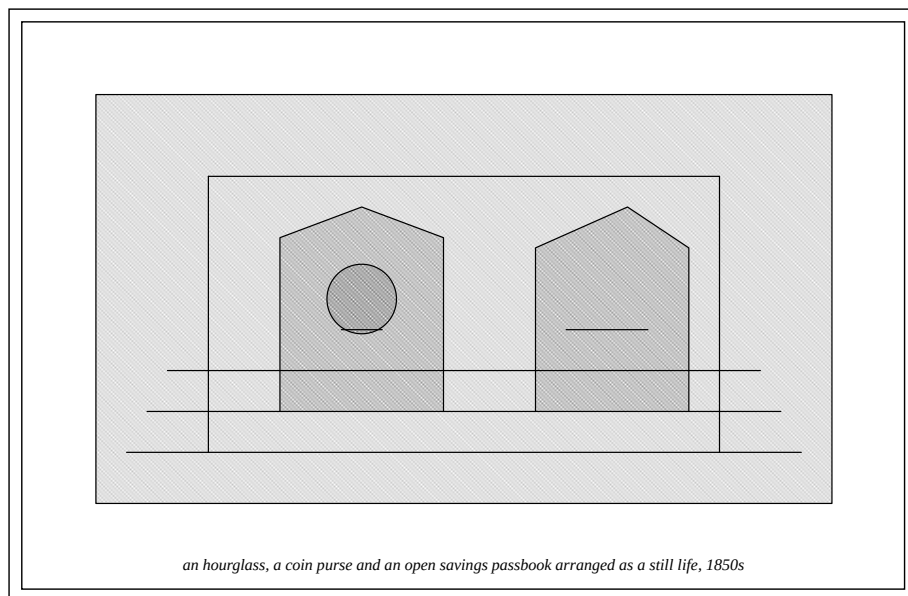
01.T1 say in plain words what finance covers and how it differs from accounting and from management

01.T2 tell profit from cash flow using a new set of sales and collection figures

01.T3 compare sole proprietorship, partnership and company for a new tailoring shop

PAUSE AND CHECK 01.P1 After the finance definition, state the rule in your own words.

PAUSE AND CHECK 01.P2 After profit versus cash flow, state the rule in your own words.



CHAPTER IV

Interest and Future Value

A practical question about interest and future value



This chapter introduces interest and future value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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In this chapter, **Interest** means The payment a borrower makes to a lender for the use of money, or the reward a saver earns. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Interest rate** means The interest for one period written as a percentage of the sum, for example 8% a year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why time changes the value of money, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

4.2 SIMPLE AND COMPOUND INTEREST

The purpose here is to make simple and compound interest concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Principal sum** means The original amount lent, borrowed or deposited, before any interest is added. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Simple interest** means Interest calculated only on the principal sum, so it is the same every period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns..

The term interest gives the chapter a precise name for this part of the problem.

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year..

The term interest rate gives the chapter a precise name for this part of the problem.

The central question in why time changes the value of money is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of simple and compound interest before moving to application.

Principal sum. The original amount lent, borrowed or deposited, before any interest is added..

The term principal sum gives the chapter a precise name for this part of the problem.

Simple interest. Interest calculated only on the principal sum, so it is the same every period..

The term simple interest gives the chapter a precise name for this part of the problem.

The business meaning matters as much as the arithmetic.

This section builds the idea of cash flows from an investment before moving to application.

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap..

The term terminal cash flow gives the chapter a precise name for this part of the problem.

Depreciation. The share of an asset's cost charged against income each year as the asset wears out..

The term depreciation gives the chapter a precise name for this part of the problem.

The central question in cash flows from an investment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the steps of the process, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

7.3 CASH FLOWS FROM AN INVESTMENT

The purpose here is to make cash flows from an investment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Terminal cash flow** means The cash received or paid when a project ends, such as the sale of scrap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Depreciation** means The share of an asset's cost charged against income each year as the asset wears out. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cash flows from an investment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to capital budgeting: process and cash flows. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep

In this chapter, **Factor table** means A table of ready-made multipliers that saves repeating a long calculation. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For future value of a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to interest and future value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times (1 + \text{rate})$. Steps: Tk 100,000 $\times (1 + 0.10) =$ Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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Factor table. A table of ready-made multipliers that saves repeating a long calculation..

The term factor table gives the chapter a precise name for this part of the problem.

The central question in future value of a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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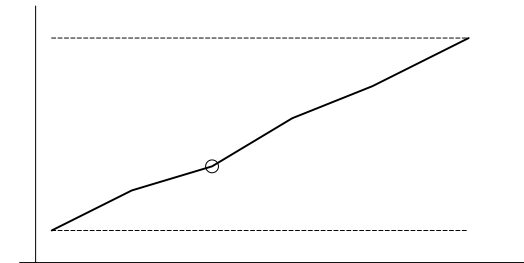


FIG. 7.1. Capital Budgeting: Process and Cash Flows figure 1

7.1 WHY FIRMS BUDGET FOR CAPITAL

This section builds the idea of why firms budget for capital before moving to application.

Capital budgeting. Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring..

The term capital budgeting gives the chapter a precise name for this part of the problem.

Capital expenditure. Spending on an asset that will help a business for more than one year..

The term capital expenditure gives the chapter a precise name for this part of the problem.

Independent project. A project that can be accepted or rejected without affecting any other project..

The term independent project gives the chapter a precise name for this part of the problem.

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other..

The term mutually exclusive projects gives the chapter a precise name for this part of the problem.

The central question in why firms budget for capital is what changes when the underlying decision changes.

The purpose here is to make why firms budget for capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital budgeting** means Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital expenditure** means Spending on an asset that will help a business for more than one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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In this chapter, **Mutually exclusive projects** means Projects of which only one can be chosen, because choosing one rules out the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

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SUMMARY

The central purpose of Interest and Future Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Annuities is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Johan de Witt, 1671 report to the States of Holland on valuing life annuities for the Dutch Republic. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

06.T1 say what makes a series of payments an annuity and name its two timing forms

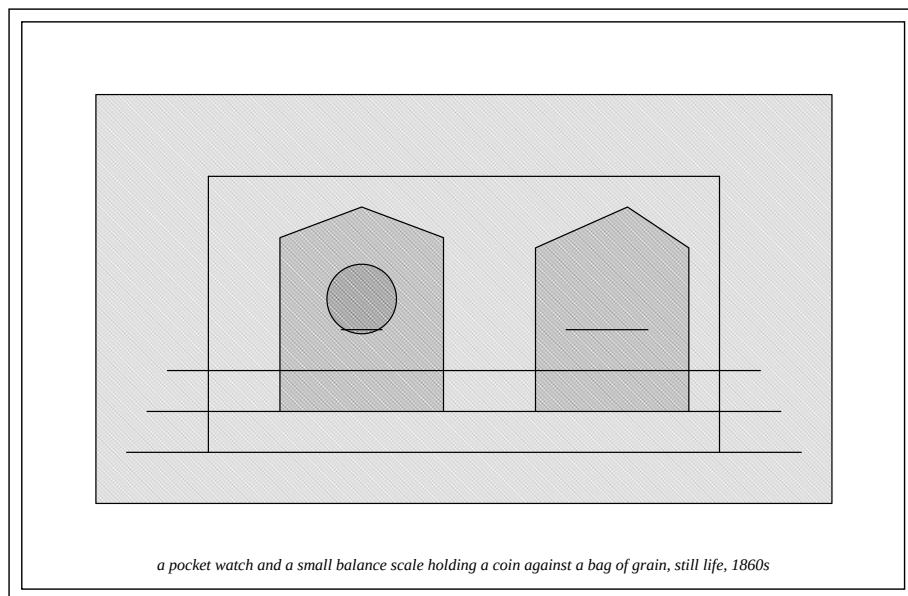
06.T2 find the future value of an annuity for several rate and period pairs as ordinary and as due

06.T3 find the present value of a new instalment plan

06.T4 compute the yearly deposit needed for a stated goal

PAUSE AND CHECK 06.P1 After annuity types, state the rule in your own words.

PAUSE AND CHECK 06.P2 After future value of an ordinary annuity, state the rule in your own words.



CHAPTER V

Present Value and Discounting

A practical question about present value and discounting



This chapter introduces present value and discounting for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

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Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For discounting a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

5.2 MIXED STREAMS OF CASH FLOWS

The purpose here is to make mixed streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Discount rate** means The yearly percentage used in discounting; it states the return you could earn elsewhere. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash inflow** means Money coming into a business or to an investor, shown as a positive number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For mixed streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in discounting a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of mixed streams of cash flows before moving to application.

Discount rate. The yearly percentage used in discounting; it states the return you could earn elsewhere..

The term discount rate gives the chapter a precise name for this part of the problem.

Cash inflow. Money coming into a business or to an investor, shown as a positive number..

The term cash inflow gives the chapter a precise name for this part of the problem.

The central question in mixed streams of cash flows is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Annuity due. An annuity whose payments fall at the start of each period..

The term annuity due gives the chapter a precise name for this part of the problem.

The central question in present value of an annuity is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of solving for the payment before moving to application.

Instalment. One of a series of regular payments used to repay a loan or build a sum..

The term instalment gives the chapter a precise name for this part of the problem.

The central question in solving for the payment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Annuity due** means An annuity whose payments fall at the start of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For present value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.4 SOLVING FOR THE PAYMENT

The purpose here is to make solving for the payment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Instalment** means One of a series of regular payments used to repay a loan or build a sum. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For solving for the payment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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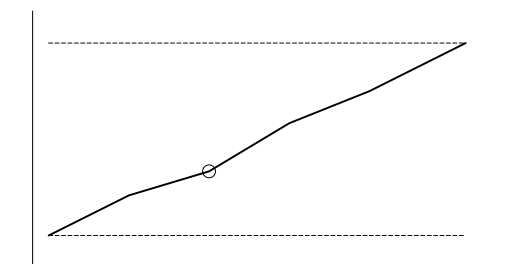


FIG. 6.1. Annuities figure 1

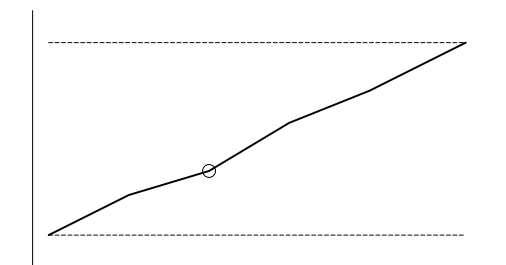


FIG. 6.2. Annuities figure 2

6.1 WHAT AN ANNUITY IS

The purpose here is to make what an annuity is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Annuity** means A series of equal payments made at equal intervals for a fixed number of periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

This section builds the idea of what an annuity is before moving to application.

Annuity. A series of equal payments made at equal intervals for a fixed number of periods..

The term annuity gives the chapter a precise name for this part of the problem.

The central question in what an annuity is is what changes when the underlying decision changes.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Present Value and Discounting is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

Irving Fisher, *The Theory of Interest* (1930), which states the discounting idea in its modern form. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

05.T1 explain discounting as the reverse of compounding using new figures

05.T2 find the present value of a single sum for two discount rates

05.T3 find present value of a mixed stream and compare two streams

PAUSE AND CHECK 05.P1 After present value of a single sum, state the rule in your own words.

PAUSE AND CHECK 05.P2 After the mixed stream, state the rule in your own words.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the capital asset pricing model before moving to application.

Unsystematic risk. Risk that affects one firm or industry only; diversification can remove it..

The term unsystematic risk gives the chapter a precise name for this part of the problem.

Beta. A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market..

The term beta gives the chapter a precise name for this part of the problem.

Market return. The average return of all investments in a market during a period..

The term market return gives the chapter a precise name for this part of the problem.

The central question in the capital asset pricing model is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For diversification and beta, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.3 THE CAPITAL ASSET PRICING MODEL

The purpose here is to make the capital asset pricing model concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Unsystematic risk** means Risk that affects one firm or industry only; diversification can remove it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

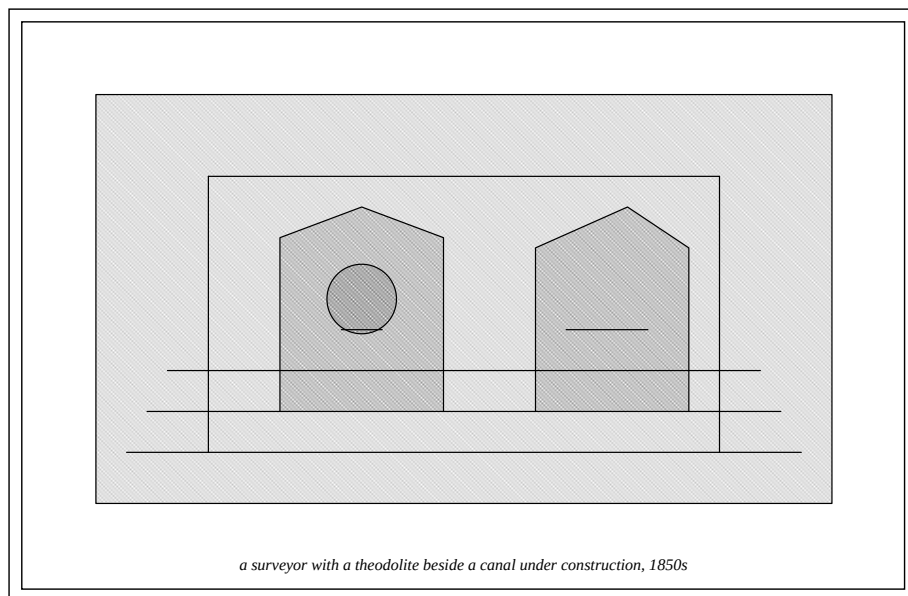
In this chapter, **Beta** means A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Market return** means The average return of all investments in a market during a period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the capital asset pricing model, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a



CHAPTER VIII

Evaluating Projects: Payback and Net Present Value

A practical question about evaluating projects: payback and net present value



This chapter introduces evaluating projects: payback and net present value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For payback period, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

8.2 NET PRESENT VALUE

The purpose here is to make net present value concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Net present value** means The present value of all a project's future cash flows minus its initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Internal rate of return** means The discount rate at which a project's net present value is exactly zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For net present value, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in payback period is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of net present value before moving to application.

Net present value. The present value of all a project's future cash flows minus its initial investment..

The term net present value gives the chapter a precise name for this part of the problem.

Internal rate of return. The discount rate at which a project's net present value is exactly zero..

The term internal rate of return gives the chapter a precise name for this part of the problem.

The central question in net present value is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

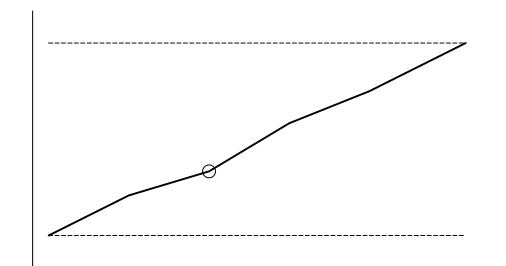


FIG. 11.1. Required Return, CAPM and Market Efficiency figure 1

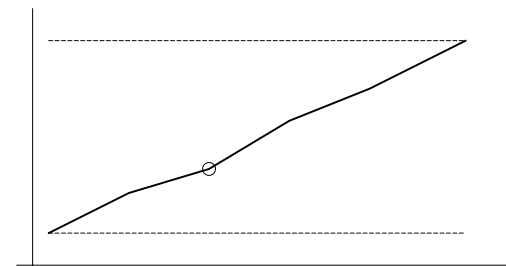


FIG. 11.2. Required Return, CAPM and Market Efficiency figure 2

11.1 RISK PREMIUM AND REQUIRED RETURN

The purpose here is to make risk premium and required return concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Required return** means The least return an investor will accept for taking a given risk. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk-free rate** means The return on an investment with no risk of loss, such as a short-term government security. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of risk premium and required return before moving to application.

Required return. The least return an investor will accept for taking a given risk..

The term required return gives the chapter a precise name for this part of the problem.

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security..

The term risk-free rate gives the chapter a precise name for this part of the problem.

to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to evaluating projects: payback and net present value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

Long-Term Capital Management, whose heavy borrowing led to a rescue organised by the Federal Reserve Bank of New York on 23 September 1998. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

10.T1 say what leverage means and tell its three kinds apart with new data

10.T2 find the break-even point and degrees of operating, financial and total leverage for a new firm

10.T3 explain in plain words why high leverage raises risk

PAUSE AND CHECK 10.P1 After break-even point, state the rule in your own words.

PAUSE AND CHECK 10.P2 After degree of operating leverage, state the rule in your own words.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Evaluating Projects: Payback and Net Present Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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The chapter has connected its main terms, method, and business decision.

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Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

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A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of financial leverage before moving to application.

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales.

The term operating leverage gives the chapter a precise name for this part of the problem.

Financial leverage. The effect of fixed financing payments on how owners' earnings respond to EBIT.

The term financial leverage gives the chapter a precise name for this part of the problem.

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales.

The term total leverage gives the chapter a precise name for this part of the problem.

The central question in financial leverage is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For operating leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.3 FINANCIAL LEVERAGE

The purpose here is to make financial leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating leverage** means The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the percentage change in EBIT divided by the percentage change in sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial leverage** means The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Total leverage** means The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For financial leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a

CHAPTER IX

Risk and Return

A practical question about risk and return



This chapter introduces risk and return for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

In this chapter, **Risk** means The chance that the actual result will differ from the result you expect. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Business risk** means Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For return and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

9.2 DESCRIBING UNCERTAINTY

The purpose here is to make describing uncertainty concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial risk** means Extra risk that comes from borrowing, because interest must be paid whatever the sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Probability distribution** means A list of possible results with the chance of each, the chances adding to one. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Risk. The chance that the actual result will differ from the result you expect..

The term risk gives the chapter a precise name for this part of the problem.

Business risk. Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing..

The term business risk gives the chapter a precise name for this part of the problem.

The central question in return and risk is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of describing uncertainty before moving to application.

Financial risk. Extra risk that comes from borrowing, because interest must be paid whatever the sales..

The term financial risk gives the chapter a precise name for this part of the problem.

Probability distribution. A list of possible results with the chance of each, the chances adding to one..

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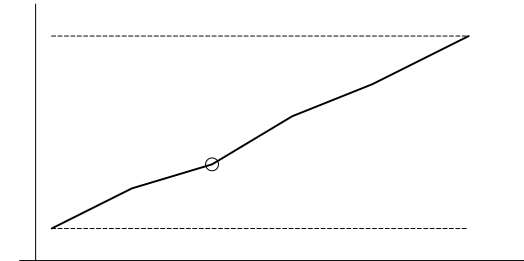


FIG. 10.1. Leverage figure 1

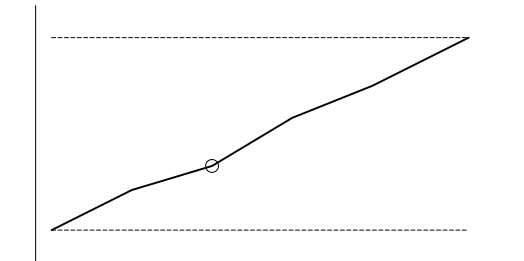


FIG. 10.2. Leverage figure 2

10.1 COSTS AND THE BREAK-EVEN POINT

The purpose here is to make costs and the break-even point concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Leverage** means The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Fixed cost** means A cost that stays the same however many units are sold, such as shop rent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of costs and the break-even point before moving to application.

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings..

The term leverage gives the chapter a precise name for this part of the problem.

Fixed cost. A cost that stays the same however many units are sold, such as shop rent..

The term fixed cost gives the chapter a precise name for this part of the problem.

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For measuring risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to risk and return. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Risk and Return is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Daniel Bernoulli, 1738 paper on measuring risk, which showed that people judge a gamble by its meaning to them as well as by its average. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

09.T1 define risk and say how business risk differs from financial risk, using a new firm

09.T2 compute the expected value, then the variance, the standard deviation and the coefficient of variation, for a new two-asset table and state which is riskier per unit of return

09.T3 explain standard deviation in plain words

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Required Return, CAPM and Market Efficiency is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

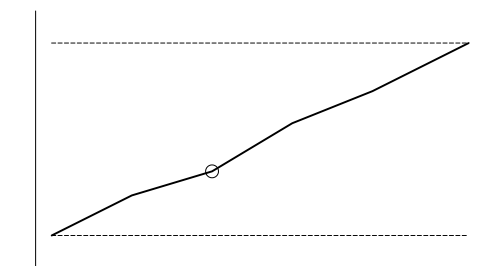


FIG. 15.1. Long-Term Financing and the Capital Market figure 1

15.1 TERM LOANS AND THE FEATURES OF FINANCING

The purpose here is to make term loans and the features of financing concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Term loan** means A loan repaid in regular instalments over a fixed number of years, usually from a bank. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Collateral** means An asset a lender may take if the borrower fails to repay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Covenant** means A condition in a loan agreement that the borrower promises to keep. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

This section builds the idea of term loans and the features of financing before moving to application.

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank..

The term term loan gives the chapter a precise name for this part of the problem.

Collateral. An asset a lender may take if the borrower fails to repay..

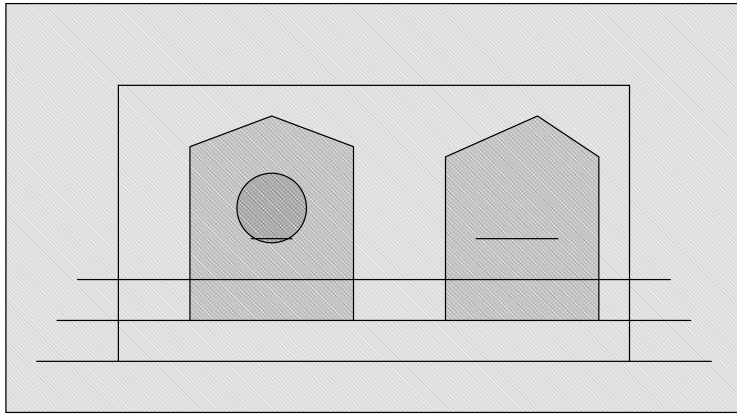
The term collateral gives the chapter a precise name for this part of the problem.

Covenant. A condition in a loan agreement that the borrower promises to keep..

The term covenant gives the chapter a precise name for this part of the problem.

The central question in term loans and the features of financing is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.



a bundle of bonds tied with ribbon and a wax seal on a lawyer's desk, 1880s

CHAPTER XII

Valuation of Bonds

A practical question about valuation of bonds



This chapter introduces valuation of bonds for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

In this chapter, **Bond** means A loan made to a company or government, repaid on a set date, with interest paid on the way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Face value** means The amount a bond repays at its end, for example Tk 1,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the idea of valuation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.2 WHAT A BOND IS

The purpose here is to make what a bond is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Coupon rate** means The yearly interest of a bond, as a percentage of its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Maturity** means The date on which a bond or loan must be repaid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Yield to maturity** means The yearly return an investor earns by buying a bond at today's price and holding it to maturity. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Bond. A loan made to a company or government, repaid on a set date, with interest paid on the way..

The term bond gives the chapter a precise name for this part of the problem.

Face value. The amount a bond repays at its end, for example Tk 1,000..

The term face value gives the chapter a precise name for this part of the problem.

The central question in the idea of valuation is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of what a bond is before moving to application.

Coupon rate. The yearly interest of a bond, as a percentage of its face value..

The term coupon rate gives the chapter a precise name for this part of the problem.

Maturity. The date on which a bond or loan must be repaid..

The term maturity gives the chapter a precise name for this part of the problem.

Yield to maturity. The yearly return an investor earns by buying a bond at today's price and holding it to maturity..

The term yield to maturity gives the chapter a precise name for this part of the problem.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Working Capital and Short-Term Financing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Toyota's just-in-time system, developed from the late 1940s by Taiichi Ohno and others to cut stock held. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 14.T1 say what working capital is and why a profitable firm can still run short
- 14.T2 compute the cost of giving up a cash discount for new terms and recommend the cheaper source
- 14.T3 find the cash conversion cycle and the financing need for new data
- 14.T4 define float and its parts in plain words

finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.4 YIELD AND PRICE

The purpose here is to make yield and price concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For yield and price, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of bonds. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

This section builds the idea of yield and price before moving to application.

The central question in yield and price is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

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A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Valuation of Bonds is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Long-term financing. Money raised for more than about five years, or with no fixed repayment date..

The term long-term financing gives the chapter a precise name for this part of the problem.

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery..

The term trade credit gives the chapter a precise name for this part of the problem.

Cash discount. A price cut a supplier gives for paying early, such as 2% off if paid within 10 days..

The term cash discount gives the chapter a precise name for this part of the problem.

Inventory. The goods a business holds for sale, such as the books on a shop's shelves..

The term inventory gives the chapter a precise name for this part of the problem.

Accounts receivable. Money that customers owe a business for goods already delivered..

The term accounts receivable gives the chapter a precise name for this part of the problem.

The central question in sources of short-term finance is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of cycles, float and estimating needs before moving to application.

In this chapter, **Long-term financing** means Money raised for more than about five years, or with no fixed repayment date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Trade credit** means Credit from a supplier who lets a buyer pay some days after delivery. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash discount** means A price cut a supplier gives for paying early, such as 2% off if paid within 10 days. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Inventory** means The goods a business holds for sale, such as the books on a shop's shelves. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Accounts receivable** means Money that customers owe a business for goods already delivered. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

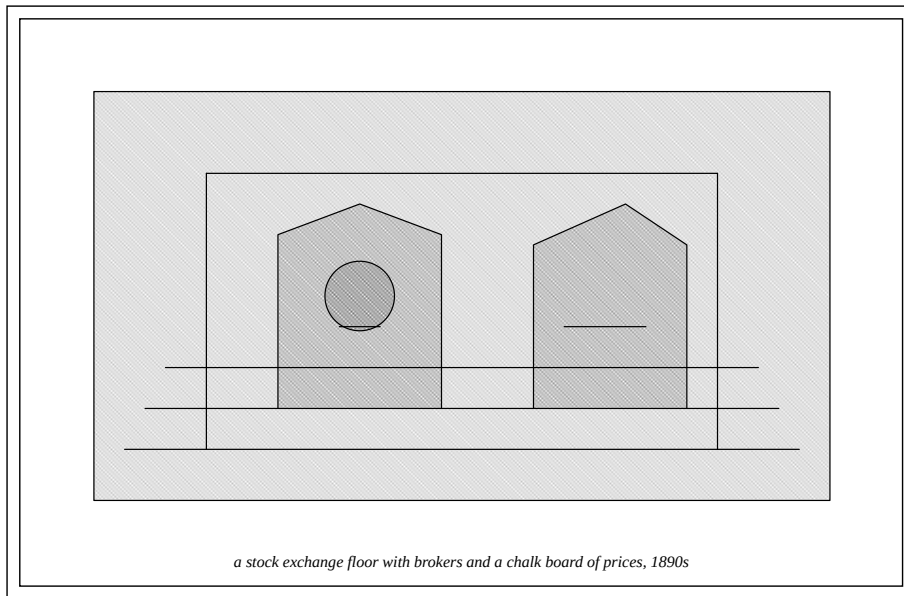
Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For sources of short-term finance, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

14.3 CYCLES, FLOAT AND ESTIMATING NEEDS

The purpose here is to make cycles, float and estimating needs concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the



CHAPTER XIII

Valuation of Shares

A practical question about valuation of shares



This chapter introduces valuation of shares for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of working capital and how to finance it before moving to application.

Working capital. The money a business has tied up in everyday trading: cash, stock and money owed to it..

The term working capital gives the chapter a precise name for this part of the problem.

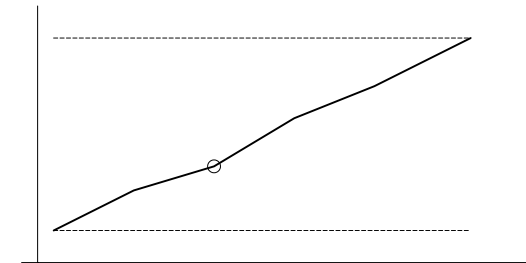


FIG. 14.1. *Working Capital and Short-Term Financing figure 1*

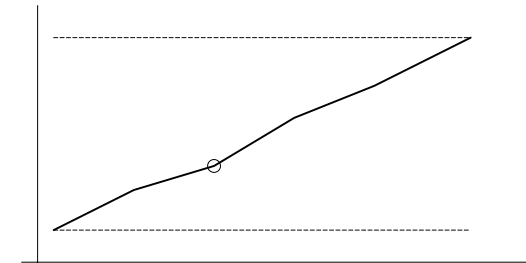


FIG. 14.2. *Working Capital and Short-Term Financing figure 2*

14.1 WORKING CAPITAL AND HOW TO FINANCE IT

The purpose here is to make working capital and how to finance it concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Working capital** means The money a business has tied up in everyday trading: cash, stock and money owed to it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

* * *

13.2 VALUING PREFERRED STOCK

The purpose here is to make valuing preferred stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital gain** means The profit from selling an asset for more than its purchase price. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Perpetuity** means A series of equal payments that continues for ever. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing preferred stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

13.3 VALUING COMMON STOCK

The purpose here is to make valuing common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend valuation model** means A way of valuing a share as the present value of the dividends it is expected to pay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of valuing preferred stock before moving to application.

Capital gain. The profit from selling an asset for more than its purchase price..

The term capital gain gives the chapter a precise name for this part of the problem.

Perpetuity. A series of equal payments that continues for ever..

The term perpetuity gives the chapter a precise name for this part of the problem.

The central question in valuing preferred stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of valuing common stock before moving to application.

Dividend valuation model. A way of valuing a share as the present value of the dividends it is expected to pay..

The term dividend valuation model gives the chapter a precise name for this part of the problem.

13.T2 value a preferred stock and a growing common stock from new data

13.T3 explain why growth raises value but also needs a required return above growth

PAUSE AND CHECK 13.P1 After common stock, state the rule in your own words.

PAUSE AND CHECK 13.P2 After perpetuity, state the rule in your own words.

In this chapter, **Cost of capital** means The return a firm must earn on its investments to satisfy those who supply its money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Weighted average cost of capital** means The average cost of a firm's different sources of money, each weighted by its share of the total. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the cost of capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to long-term financing and the capital market. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

Cost of capital. The return a firm must earn on its investments to satisfy those who supply its money..

The term cost of capital gives the chapter a precise name for this part of the problem.

Weighted average cost of capital. The average cost of a firm's different sources of money, each weighted by its share of the total..

The term weighted average cost of capital gives the chapter a precise name for this part of the problem.

The central question in the cost of capital is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

reasoning should use the facts given in the question and keep timing and units consistent.

13.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

13.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XIV · WORKING CAPITAL AND SHORT-TERM FINANCING

14.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

14.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

14.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XV · LONG-TERM FINANCING AND THE CAPITAL MARKET

15.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Long-Term Financing and the Capital Market is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

reasoning should use the facts given in the question and keep timing and units consistent.

08.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

08.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

08.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER IX · RISK AND RETURN

09.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

09.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

09.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER X · LEVERAGE

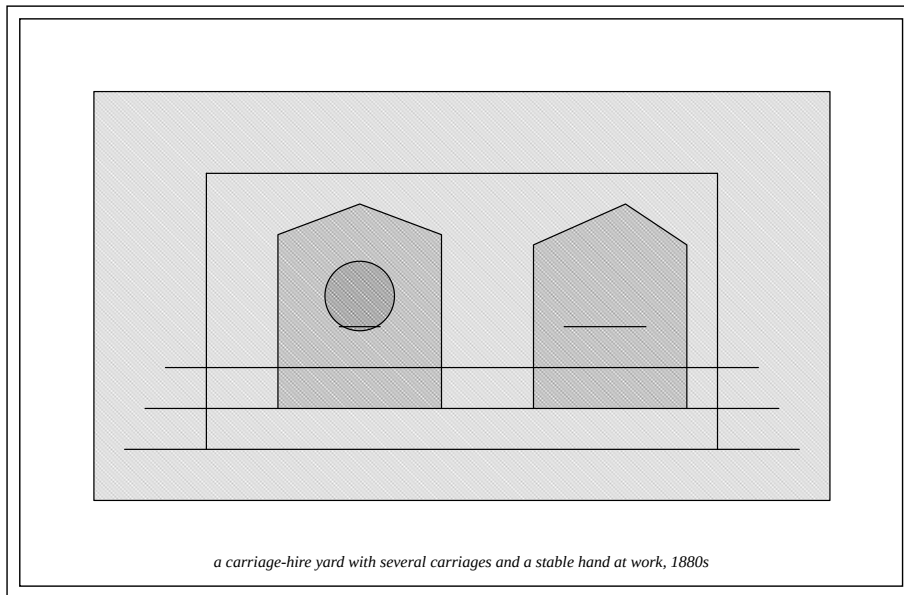
10.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

10.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

10.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.



CHAPTER XVI

Leasing

A practical question about leasing



This chapter introduces leasing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

03.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

03.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

03.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER IV · INTEREST AND FUTURE VALUE

04.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

04.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER V · PRESENT VALUE AND DISCOUNTING

05.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

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CHAPTER VI · ANNUITIES

06.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.2 KINDS OF LEASE

The purpose here is to make kinds of lease concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating lease** means A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial lease** means A long lease, not cancellable, that covers most of the asset's life and works like a loan. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sale-and-leaseback** means A deal in which a firm sells an asset it owns and leases it back from the buyer. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For kinds of lease, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.3 THE LEASE-OR-BUY DECISION

The purpose here is to make the lease-or-buy decision concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

This section builds the idea of kinds of lease before moving to application.

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset..

The term operating lease gives the chapter a precise name for this part of the problem.

Financial lease. A long lease, not cancellable, that covers most of the asset's life and works like a loan..

The term financial lease gives the chapter a precise name for this part of the problem.

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer..

The term sale-and-leaseback gives the chapter a precise name for this part of the problem.

The central question in kinds of lease is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the lease-or-buy decision before moving to application.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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Risk premium. The extra return demanded for taking risk instead of a risk-free investment. (*Chapter XI*)

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security. (*Chapter XI*)

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer. (*Chapter XVI*)

Security market line. The graph of the capital asset pricing model, with beta on one axis and required return on the other. (*Chapter XI*)

Semiannual interest. Interest paid twice a year, each payment being half the yearly amount. (*Chapter XII*)

Share price. The price at which one share of a company can be bought or sold today. (*Chapter III*)

Shareholder. A person who owns part of a company by holding its shares. (*Chapter III*)

Short-term financing. Money borrowed to be repaid within one year. (*Chapter XIV*)

Simple interest. Interest calculated only on the principal sum, so it is the same every period. (*Chapter IV*)

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts. (*Chapter I*)

Stakeholder. Anyone affected by a firm's actions, such as staff, customers, lenders and the community. (*Chapter III*)

Standard deviation. The square root of the variance; it shows how far results typically stray from the expected value. (*Chapter IX*)

Sunk cost. Money already spent that cannot be recovered whatever you decide now. (*Chapter VII*)

Systematic risk. Risk that affects all investments, such as a recession; it cannot be diversified away. (*Chapter XI*)

Tax shield. The tax saved because a cost, such as interest or depreciation, reduces taxable income. (*Chapter XVI*)

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank. (*Chapter XV*)

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap. (*Chapter VII*)

Time value of money. The idea that a taka received today is worth more than a taka received later. (*Chapter IV*)

Timeline. A line marking each period, with the cash flow written at the point where it occurs. (*Chapter IV*)

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. (*Chapter X*)

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery. (*Chapter XIV*)

Treasurer. The officer in charge of handling cash, raising money and keeping relations with banks. (*Chapter II*)

what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

United States Leasing Corporation, founded in San Francisco in 1952, an early firm of the modern equipment-leasing industry. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

16.T1 say what a lease is and compare operating and financial leases with a new asset

16.T2 list the steps of the lease-or-buy comparison

16.T3 compute the after-tax present value of a new lease and purchase

16.T4 explain the tax shield of interest and depreciation

PAUSE AND CHECK 16.P1 After kinds of lease, state the rule in your own words.

PAUSE AND CHECK 16.P2 After after-tax outflows, state the rule in your own words.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns. (*Chapter IV*)

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year. (*Chapter IV*)

Intermediate-term financing. Money borrowed to be repaid in more than one year but not more than about five. (*Chapter XIV*)

Internal rate of return. The discount rate at which a project's net present value is exactly zero. (*Chapter VIII*)

Inventory. The goods a business holds for sale, such as the books on a shop's shelves. (*Chapter XIV*)

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. (*Chapter II*)

Lease. A contract in which the owner of an asset lets another party use it for regular payments. (*Chapter XVI*)

Lessee. The user who pays to use the asset under a lease. (*Chapter XVI*)

Lessor. The owner who lets the asset out under a lease. (*Chapter XVI*)

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. (*Chapter X*)

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill. (*Chapter II*)

Long-term financing. Money raised for more than about five years, or with no fixed repayment date. (*Chapter XIV*)

Market return. The average return of all investments in a market during a period. (*Chapter XI*)

Matching principle. Pay for short-lived needs with short-term money and for lasting needs with long-term money. (*Chapter XIV*)

Maturity. The date on which a bond or loan must be repaid. (*Chapter XII*)

Mixed stream. A series of cash flows of different sizes in different periods. (*Chapter V*)

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other. (*Chapter VII*)

Net present value. The present value of all a project's future cash flows minus its initial investment. (*Chapter VIII*)

Net working capital. Current assets minus current liabilities; what remains for trading after short debts are paid. (*Chapter XIV*)

Operating cash flow. The cash a project brings in each year after paying its cash costs and tax. (*Chapter VII*)

Operating cycle. The days from buying stock to collecting cash from its sale. (*Chapter XIV*)

Operating efficiency. How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. (*Chapter I*)

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. (*Chapter XVI*)

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the

APPENDIX A

Maths You Will Use



- Cash inflow.** Money coming into a business or to an investor, shown as a positive number. (*Chapter V*)
- Cash outflow.** Money going out of a business or from an investor, shown as a negative number. (*Chapter V*)
- Coefficient of variation.** Standard deviation divided by expected value, giving risk per unit of expected return. (*Chapter IX*)
- Collateral.** An asset a lender may take if the borrower fails to repay. (*Chapter XV*)
- Common stock.** Shares that give owners a vote and a claim on profit after all other claims are met. (*Chapter XIII*)
- Company.** A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. (*Chapter I*)
- Compound interest.** Interest calculated on the principal sum and also on interest already earned. (*Chapter IV*)
- Compounding period.** The length of time after which interest is added to the balance, such as a year or six months. (*Chapter IV*)
- Constant growth model.** The dividend valuation model for dividends that grow at the same rate every year. (*Chapter XIII*)
- Contribution margin.** Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. (*Chapter X*)
- Controller.** The officer in charge of accounting records, reports and checking that money is used as planned. (*Chapter II*)
- Cost of capital.** The return a firm must earn on its investments to satisfy those who supply its money. (*Chapter XV*)
- Coupon rate.** The yearly interest of a bond, as a percentage of its face value. (*Chapter XII*)
- Covenant.** A condition in a loan agreement that the borrower promises to keep. (*Chapter XV*)
- Debenture.** A bond backed only by the issuer's general promise to pay, not by a particular asset. (*Chapter XV*)
- Depreciation.** The share of an asset's cost charged against income each year as the asset wears out. (*Chapter VII*)
- Discount bond.** A bond that sells for less than its face value. (*Chapter XII*)
- Discount rate.** The yearly percentage used in discounting; it states the return you could earn elsewhere. (*Chapter V*)
- Discounted payback period.** The payback time when each cash flow is first discounted to present value. (*Chapter VIII*)
- Discounting.** Finding present value by removing the interest a sum could earn before it arrives. (*Chapter V*)
- Diversification.** Spreading money across many investments so that one failure does less damage. (*Chapter XI*)
- Dividend.** A share of a company's profit paid out to its owners instead of being kept in the business. (*Chapter II*)
- Dividend decision.** A choice about how much profit to pay out to owners and how much to keep for the business. (*Chapter II*)

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